Semiempirical Orbital-Optimized DFT in MOUSE: Spin-Pure Excited-State Surfaces with Analytic Gradients and Hessians

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Working manuscript, 1 November 2026

Abstract

We develop a semiempirical orbital-optimized route to excited-state energies, gradients and Hessians in MOUSE, motivated by variational electronic saddle searches. A spin-polarized GFN2 energy uses one-time DFT-derived orbital references, fixed 50/50 F/Q weights, exchange-response scale 0.9, and zero excitation-energy offset. Spin-pure optimized singlets couple to selected double excitations through a prepared static screened interaction with one calibrated bath efficiency, $\eta = 0.60$. A spectral-block construction treats native orbitals whose full-DFT weight is fragmented over several poles. On 26 exposed development states in 24 molecules, the resulting candidate gives 0.198 eV mean absolute error and 0.446 eV maximum error. Twelve $\pi\pi^*$ states give 0.216 eV MAE, compared with 1.514 eV for a matched TDA control on the same corrected native GFN2 orbitals. For fourteen $n\pi^*$ states, the respective MAEs are 0.183 and 0.168 eV. The spectral-block amendment was developed after pyrrole scoring; these results are not blind validation. Complete analytic response includes orbital stationarity, spin recoupling, the configuration metric, the moving ground projector and state-eigenvector response. The prepared MOUSE/LINK surface follows coherent singlets without a runtime TDA solve, new SCC call or DFT-backend import. Independent scalar-energy checks verify the analytic derivatives; ethylene's full Hessian takes 10.68 s in one two-thread run. Earlier-model minimum and frequency comparisons are identified separately. A larger carotene vertical example is calculated and pigment structures are prepared; these remain excluded from accuracy claims. Independent transferability, equilibrium-structure and complete-cost benchmarks remain necessary.

1 Introduction

The cost of an excited-state calculation is set by the scientific task, not only by the cost of a single vertical energy. Conformer and solvent ensembles, excited-state geometry searches, and local vibrational analyses require repeated evaluations on related molecular structures. A compact, parameterized electronic representation can make these repeated tasks accessible while retaining explicit orbitals, charge redistribution, and a differentiable energy. The purpose of a semiempirical model is therefore to allocate a finite computational budget across the structures and derivatives needed for a molecular prediction. It is not to replace high-level reference calculations indiscriminately. Its usefulness requires a controlled balance of electronic error, nuclear sampling, and computational cost.

The simplified excited-state methods of Grimme and co-workers provide a major reference point. The original sTDA approximates the response matrix and screens the excitation space while retaining a DFT ground-state reference; sTD-DFT extends the simplified construction beyond the Tamm–Dancoff approximation.[1, 2] These are approximations to an excited-state response problem and need not imply a semiempirical ground-state Hamiltonian. In sTDA-xTB, a dedicated tight-binding Hamiltonian and an augmented basis replace the expensive ground-state

reference, enabling a predominantly semiempirical spectroscopy workflow.[3] That reference is not interchangeable with an ordinary GFN2 ground-state calculation: the description of the virtual space and the response parameterization are part of the method.

This family has continued to develop. The 2026 std2 description includes sTDA, sTD-DFT, their exact-integral simplified variants XsTDA and XsTD-DFT, and spin-flip formulations, with applications to a broad range of optical response properties.[4] These developments make simplified response methods strong alternatives for multi-root spectroscopy and realistic molecular environments. The present work addresses a complementary objective: an explicitly optimized semiempirical excited-state energy and its analytic nuclear curvature. Its matched GFN2 TDA controls isolate this choice within the MOUSE energy models. Earlier model-generation comparisons with published sTDA-xTB are retained in the Supporting Information; XsTD-DFT is not benchmarked here.

Other low-cost approaches already address excited-state structure and dynamics. TD-DFTB and its long-range-corrected and hybrid extensions combine a compact DFT-derived Hamiltonian with an excited-state response treatment. Recent hybrid-functional DFTB parameterizations target organic photovoltaic molecules, illustrating that frontier levels and charge transfer remain active areas of model development.[5, 6] TDDFT+TB instead keeps a DFT ground state and simplifies the coupling matrix; analytic excited-state gradients were reported in 2023.[7] The approximate auxiliary-function TDDFT-aas approach added an alternative efficient response construction with analytic gradients in 2024.[8] These DFT-reference approaches are relevant cost-reduction strategies, although they do not make the entire calculation semiempirical.

A particularly close precedent is ΔDFTB. Kowalczyk, Le, and Irle implemented state-specific excited-state energies and gradients within DFTB, with benchmarks of geometries, Stokes shifts, and vibrational frequencies.[9] The method remains part of the modern DFTB+ framework.[5] A direct semiempirical excited-state energy is therefore an established idea, not a first introduced here. Orthogonalization-corrected OM2/OM3 and ODM2/ODM3 Hamiltonians combined with multireference configuration interaction (MRCI) provide a different established route to excited-state energies and molecular structure, with a wavefunction expansion rather than a single targeted orbital-optimized CSF.[10, 11] Their treatment of several configurations is especially relevant when evaluating the eventual scope of a ROKS-based model.

The methodological motivation comes from the orbital-optimization studies of Levi and co-workers. Levi, Ivanov, and Jónsson formulated variational excited-state calculations as direct optimization in orbital space, including convergence to electronic saddle points rather than relying exclusively on self-consistent-field iterations.[12, 13] Their work separates the definition of a stationary electronic state from the algorithm used to locate it. Ivanov and co-workers extended this approach to grid and plane-wave representations and discussed excited-state atomic forces.[14] Generalized mode following subsequently provided a way to target a specified saddle order by reversing selected electronic curvature directions.[15] The more recent freeze-and-release strategy uses an initial relaxation with selected excitation orbitals fixed, followed by their release in a full saddle search.[16] Together, these contributions motivate an excited-state solver built around an energy, its orbital gradient, and its curvature.

This perspective is particularly attractive for a semiempirical platform. A ground-state engine already supplies much of the required infrastructure: the energy, Fock matrix, density response, overlap derivatives, and nuclear derivatives. An orbital-optimized excited-state method can reuse these quantities, provided the state-specific orbital equations and their response are added consistently. Reusing the engine therefore means reusing energy and derivative primitives; it does not make an unmodified ground-state Hessian an excited-state Hessian.

MOUSE, the Molecular Orbital Utility for Semiempirical Explorations, supplies a modular environment for such a development. Here its GFN2-based energy is augmented by frozen operators obtained from projected ground-state DFT information. GFN2-xTB provides the semiempirical foundation,[17] while restricted open-shell Kohn–Sham (ROKS) ideas provide a

spin-adapted open-shell singlet construction.[18] The resulting method is called *semiempirical orbital-optimized DFT* (OO-DFT) in the sense that a density-functional-response-informed semiempirical energy is made stationary with respect to the excited-state orbitals. It is not a full-basis Kohn–Sham calculation at each molecular geometry.

Against this background, our contribution is the combination of a GFN2-based, DFT-response-informed energy, a spin-pure ROKS CSF, direct orbital stationarity with state following, and an analytic constrained response through the nuclear Hessian. The contribution concerns this explicit formulation and implementation, rather than priority for semiempirical excited-state gradients or orbital optimization in general. The final examples address vertical accuracy, numerical correctness of the derivatives, portable state following, local anchor sensitivity and final-model molecular minima. Earlier molecular-minimum examples remain separately identified. TDA is retained as a matched control, using semiempirical rather than DFT orbitals. No general replacement of TDA for all excited-state tasks is implied.

Figure 1 summarizes the flow from frozen reference information and the native semiempirical engine to the stationary excited-state surface and its derivatives.

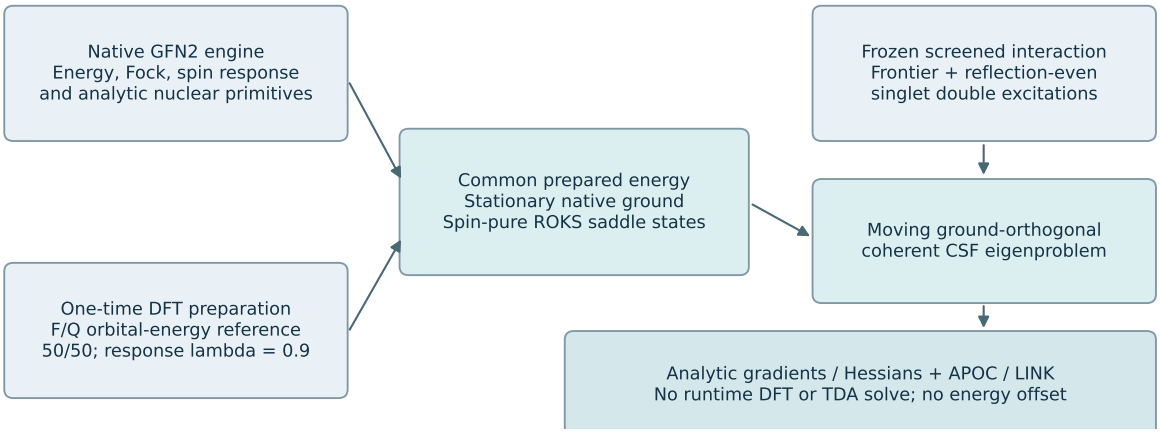


Figure 1: Architecture of the prepared MOUSE OO-CID surface. F/Q information defines one common 50/50 reference before orbital optimization. Native GFN2 energy, spin and derivative primitives support stationary ROKS states. Frozen screened couplings and symmetry-selected singlet configurations define the moving-ground-orthogonal state interaction. Full implicit and eigenvector response gives analytic nuclear derivatives; APOC/LINK retain the coherent spin-pure state.

2 Theory and implementation

2.1 Common reference and spin-polarized native energy

The native GFN2 spatial orbitals C_{n} define the retained valence chart. At the frozen anchor, a core-orthogonal isometry embeds this chart in the DFT AO space. The diagonal Rayleigh quotients of the embedded DFT Fock matrix supply levels F_i. Qualified occupied/virtual assignments transfer DFT orbital spacings to Q_i after aligning occupied centroids; unqualified levels retain F_i. The correspondence thresholds are 0.7 in absolute overlap and 0.05 in best-versus-alternative margin. If the molecular Q transfer loses its occupied–virtual gap, Q reverts to F. The common one-body reference uses $\epsilon_i = (F_i + Q_i)/2$ before orbital optimization. DFT spatial orbitals are not substituted for the native orbitals. This is not an arithmetic average of two independently optimized excited energies.

In a complete symmetric AO-pair moment representation, the prepared energy is

$$E[P^\alpha, P^\beta; R] = E_{\text{native}} + v^\mathsf{T} q + \tfrac{1}{2}(q - q_0)^\mathsf{T} K_c (q - q_0) + \tfrac{1}{2} m^\mathsf{T} K_s m, \tag{1}$$

$$q_a = \mathrm{Tr}[B_a (P^\alpha + P^\beta)], \qquad m_a = \mathrm{Tr}[B_a (P^\alpha - P^\beta)]. \tag{2}$$

E_{native} retains the native electronic, charge-dependent, anisotropic, repulsive, dispersion and spin-polarization terms; K_c and K_s are the prepared additional response corrections. The charge correction subtracts the corresponding native response rather than adding a duplicate Coulomb kernel. The linear term places the reference Fock spectrum in the native chart. The quadratic tensors are prepared at the stationary embedded DFT density and the linear reference is centered on the native ground. Spin polarization is part of the energy and its derivatives, not an additive excitation correction.

The final reference is CAM-B3LYP,[19] with range parameter $0.33\,\mathrm{bohr}^{-1}$. The inherited uracil scale $\lambda = 0.9$ multiplies its exact-exchange response once, giving short- and long-range response coefficients 0.171 and 0.585. The semilocal response is unchanged and orbital gaps are not multiplied by λ. Neither operator construction nor state assignment uses a reference excitation energy.

2.2 Spin purity and electronic saddle stationarity

For a promotion $h \to p$, the two exchanged-spin determinants sharing a single spatial orbital set define a singlet CSF with a consistent fermion phase convention. Its energy is the restricted-open-shell multiplet sum

$$E_S(C, R) = 2E_M(C, R) - E_T(C, R). \tag{3}$$

The singlet CSF is spin pure, whereas the density-functional multiplet-sum energy remains approximate. This does not identify a broken-symmetry $M_s = 0$ determinant as an exact singlet. Four-open-shell occupations require two independent singlet recouplings, both retained below.

The ground is made stationary in the native closed-shell functional. Target ROKS orbitals satisfy the ground-CSF orthogonality constraint and their bordered stationarity equations. Metric-preserving orbital rotations, Newton residual searches, and state-overlap guards locate the assigned branch. An excited solution may be an electronic saddle, as emphasized in the direct-optimization work of Levi and co-workers;[13, 15] no orbital-energy minimum is assumed. Electronic saddle order and nuclear vibrational stability are distinct properties.

2.3 Analytic response of the state-interaction surface

Let z collect the stationary ground and target orbital coordinates and constraint multipliers, and write their residual equations as $F(z(R), R) = 0$. Their first and second analytic responses obey

$$F_z z_a = -F_a, \tag{4}$$

$$F_z z_{ab} = -F_{ab} - F_{az} z_b - F_{bz} z_a - F_{zz}[z_a, z_b]. \tag{5}$$

The bordered linear systems are solved on the regular assigned branch. Both responses are required: the subsequent state-interaction eigenvalue is not stationary with respect to the individually optimized ROKS orbitals. A variational single-determinant Hessian formula alone would omit terms.

For the generalized eigenproblem in the moving ground-orthogonal space, $Hc = \mathcal{E}Sc$ with $c^\mathsf{T} S c = 1$, total matrix derivatives give

$$\mathcal{E}_a = c^\mathsf{T}(H_a - \mathcal{E}S_a)c, \tag{6}$$

$$\mathcal{E}_{ab} = c^\mathsf{T}(H_{ab} - \mathcal{E}S_{ab})c - \mathcal{E}_b c^\mathsf{T} S_a c - \mathcal{E}_a c^\mathsf{T} S_b c + 2\sum_{k \neq n} \frac{[c_k^\mathsf{T}(H_a - \mathcal{E}S_a)c][c_k^\mathsf{T}(H_b - \mathcal{E}S_b)c]}{\mathcal{E} - \mathcal{E}_k}. \tag{7}$$

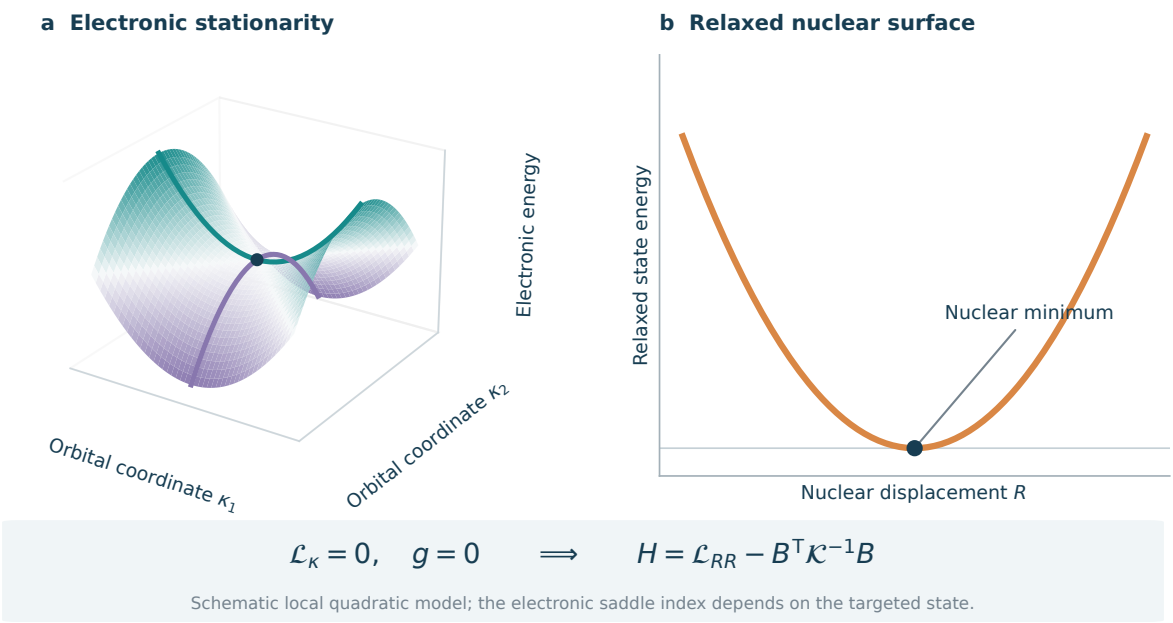


Figure 2: Schematic distinction between electronic and nuclear stationarity. The local quadratic illustration has one unstable orbital direction and a stable relaxed nuclear branch; it is not a molecular calculation. The number of unstable orbital directions need not be one. The displayed quadratic Schur-complement illustration does not replace the full CI response below.

Here the derivatives include the orbital/multiplier responses, metric, transported operators, and moving ground projector. The relevant metric and eigenvalue separations must be resolved. The absolute excited-state energy defines molecular forces and curvature; vertical excitation energies subtract the stationary native ground. Finite differences enter only the independent validation of these analytic derivatives.

3 Computational protocol

The exposed development assessment contains twenty valence transitions in eighteen molecules: seven $\pi\pi^*$ and thirteen $n\pi^*$ targets. Reference energies are the archived QUEST theoretical best estimates, with individual method labels retained in the input data.[20, 21] Molecular geometries and target identities are unchanged from the frozen inputs. All twenty numerical labels were exposed during development of the final augmentation; none is presented as an independent blind test. The separately frozen three-state prospective extension and its preparation amendment are reported below.

The prepared CAM-B3LYP references retain the archived aug-cc-pVDZ AO basis, numerical grid level 3, frozen core embedding, and native valence space. Archived DFT transition information supplies assigned-state seeds and character checks in this benchmark. It does not supply excitation-energy shifts and is not read during prepared runtime. Preparation, including this reference information, remains distinct from evaluation of the MOUSE surface. Completed references, stationary states and operators are reused.

Earlier reference calculations were performed on `oracle.sns.it`; the final configuration-space development and derivative validations were run on the separately verified MAC-STUDIO host. No production electronic calculation was run on the laptop. Runtime property checks use two BLAS threads per process. The parallel nuclear-column validation limits concurrent processes explicitly and preserves every completed column checkpoint.

Matched TDA controls use the same stationary native GFN2 spatial orbitals, F/Q one-body preparation, density-response kernel and $\lambda = 0.9$:

$$A_{ia,jb} = (\epsilon_a - \epsilon_i)\delta_{ij}\delta_{ab} + K_{ia,jb}. \tag{8}$$

The complete nonlocal singlet A operator is used, including the exchange contribution missing from a real symmetrized static response. These controls are distinct from published sTDA-xTB and from a separately optimized TDA parameterization. The screened interaction introduced below is an additional ingredient of OO-CID, not a hidden modification of the reported TDA values.

4 Screened, symmetry-augmented optimized-target CID

The final development candidate addresses the two remaining $\pi\pi^*$ outliers by changing the configuration space and its interaction, while retaining the native GFN2 orbitals, the same prepared F/Q one-body reference, 50/50 weights, $\lambda = 0.9$, and zero excitation-energy offset. It combines optimized ground-orthogonal ROKS target singlets, their paired doubles, and the original two-occupied/two-virtual frontier doubles in both reflection parities. The added configurations are all spin-singlet doubles from the five highest occupied and five lowest virtual canonical native orbitals that are even under reflection in the molecular plane. Both independent singlet recouplings of every four-open-shell occupation are included. Canonical paired doubles with retained-CSF overlap at least 0.7 are omitted by the unchanged duplicate-removal rule. An exact Gaussian-overlap representation certifies the nuclear reflection and canonical MO parities at preparation. The resulting configuration list is frozen during subsequent nuclear displacement; it is not reclassified when an out-of-plane coordinate changes. If reflection or MO parity cannot be certified, no additional symmetry-sector configuration is selected.

This is one common CSF space and one effective matrix for all requested states of a molecule. No $n\pi^*/\pi\pi^*$ label, reference excitation energy, molecular exception, or fitted damping factor enters the selection. For the planar examples, $\pi\pi^*$ targets are reflection-even, whereas $n\pi^*$ targets are reflection-odd. The symmetry augmentation therefore adds the missing even-sector flexibility without introducing the large odd-sector manifold that overcorrelated several controls in the unrestricted five-pair diagnostic. The choice was made after inspecting the exposed twenty-state development set and was frozen in commit `2d59ab8`. It is a model-development choice, not blind validation or a general prescription for nonplanar molecules, CT, or Rydberg excitations.

Environmental screening is introduced through a static finite-basis partition motivated by constrained RPA.[22] Let V be the bare Coulomb matrix in the orthonormal AO-pair chart, Δ the positive native occupied–virtual gap matrix, and T the symmetrized transition-density columns weighted by their environmental fraction. With occupied and virtual active projectors P_h, P_v, these fractions are $1 - (P_h)_{ii}(P_v)_{aa}$; the projectors span the five-pair frontier and retained optimized hole/particle directions. Our prepared static interaction is

$$W = V - VT(\Delta + T^{\mathsf{T}}VT)^{-1}T^{\mathsf{T}}V. \tag{9}$$

This explicitly specified finite-basis approximation is not a full frequency-dependent cRPA calculation. The same W acts in both reflection sectors. Its one-body term is normal ordered to the unchanged native reference Fock matrix, $h_W = F_{\mathrm{ref}} - J_W[P_g] + K_W[P_g]/2$. Let H^W_{IJ} and S_{IJ} denote its Slater–Condon matrix and CSF metric. The native spin-polarized functional supplies each diagonal energy E_I, and the effective matrix is

$$H_{IJ} = H^W_{IJ} + \tfrac{1}{2}S_{IJ}(d_I + d_J), \qquad d_I = E_I - H^W_{II}/S_{II}. \tag{10}$$

Every two-dimensional four-spin singlet block is restored to the exact SU(2) completion of the same quadratic spin functional. Excited CSFs are projected orthogonal to the moving stationary

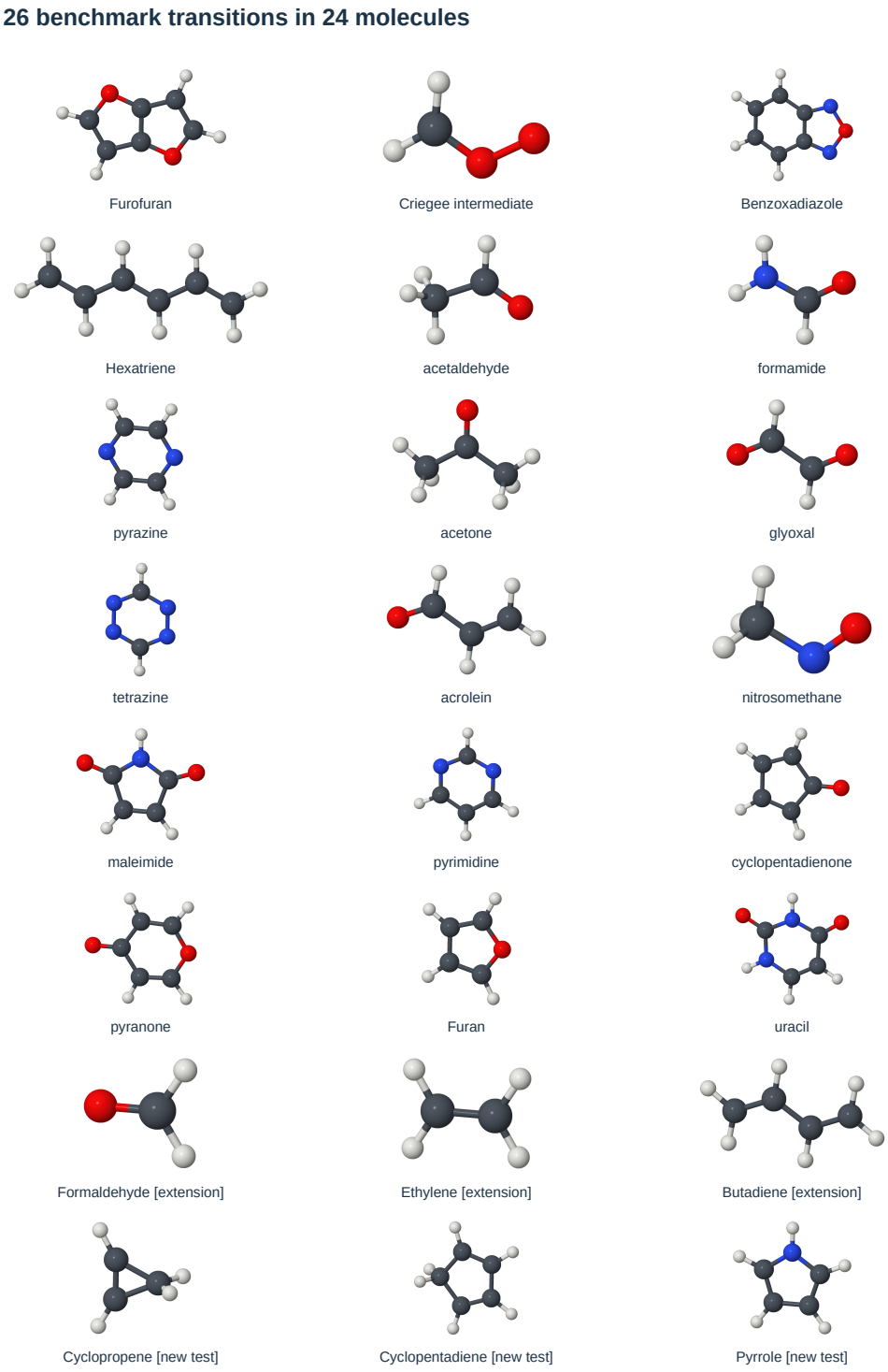


Figure 3: The twenty-four molecular anchors used for the twenty-state development set and its two three-state extensions, rendered with Jmol from the archived three-dimensional coordinates. Only rigid display rotations are applied; panel scales differ. Furofuran contributes two $\pi\pi^*$ states, and Criegee contributes one state of each valence class.

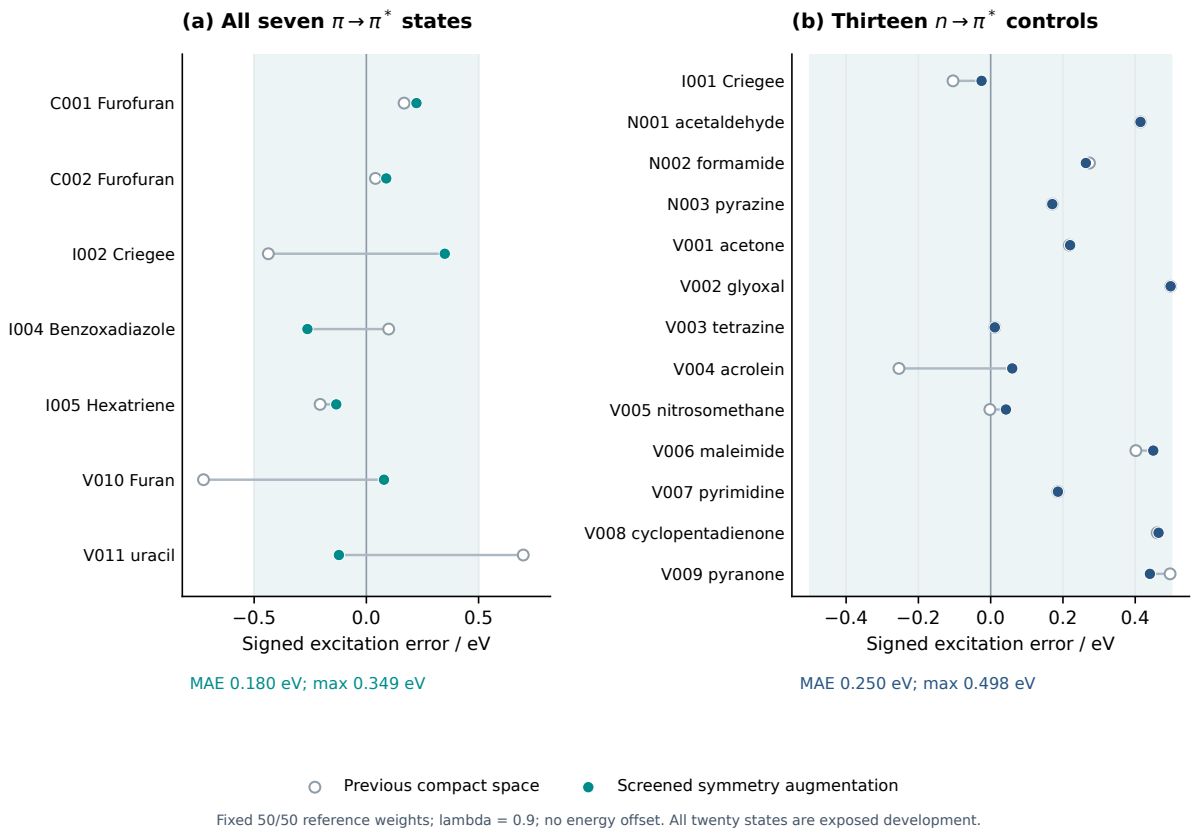


Figure 4: Resolution of the seven $\pi\pi^*$ development targets while retaining the thirteen $n\pi^*$ controls. Open circles show the previous compact-space errors; filled circles show the screened symmetry augmentation. The shaded interval is ±0.5 eV. All twenty states informed the development assessment; this figure is not a blind-validation result.

ground CSF before solving the generalized eigenproblem. Excitation energies subtract its native stationary energy, not an auxiliary correlated-ground eigenvalue. The overlap-weighted residual remains an approximate multistate functional; it is not a proof of a universally balanced correlated ground-state theory.

All seven $\pi\pi^*$ targets now satisfy the unchanged 0.5 eV individual criterion: their MAE is 0.180 eV and maximum absolute error is 0.349 eV. Furan changes from 5.804 to 6.609 eV against 6.530 eV, and uracil from 6.080 to 5.259 eV against 5.381 eV. The thirteen $n\pi^*$ controls remain within 0.5 eV, with 0.250 eV MAE and 0.498 eV maximum error. Overall, the twenty states give 0.225 eV MAE. These are exposed-data development results: the earlier post-freeze designation does not make them prospective tests of this newly selected model. The matched native-orbital TDA control retains its 0.925 eV MAE, using the same F/Q preparation and native response kernel; the new screened interaction is an additional ingredient of OO-CID. This comparison does not establish superiority to every calibrated TDA, sTDA-xTB, or TD-DFTB parameterization. All individual values are in the Supporting Information.

The public `PreparedNOSIModel` accepts a checksummed extended model, and LINK retains its `nosi_50_50` selector and accepted-restart contract. Runtime evaluation reuses only ground-engine Fock, energy, spin-polarization, and derivative primitives; it performs no TDA solve, DFT-backend import, or new SCC calculation after preparation. APOC follows the coherent determinant sum and verifies its norm and $\langle S^2 \rangle = \|S_+\Psi\|^2 = 0$. An individual broken-symmetry $M_s = 0$ determinant is never identified as a spin-pure singlet. The analytic response differentiates the native ground and target KKT systems, all diagonal functionals and spin recouplings, the CSF metric, generalized eigenvectors, and moving ground projector. The prepared h_W, W,

auxiliary orbitals, and configuration list are fixed in the co-rotating orthonormal AO frame. The resulting derivatives are exact for this anchored surface; rebuilding the DFT reference or screening at each geometry defines a different model.

Canonical auxiliary blocks are contracted exactly with frozen-core occupation bitstrings, while optimized-state cross terms retain the singular-safe nonorthogonal Schur/cofactor algebra. Further acceleration caches NumPy's greedy contraction programs, omits only exactly zero derivative terms, and reuses the invariant auxiliary Gram block. Neither an integral nor a derivative term is approximated. Independent canonical-block comparisons agree below 10^{-12} Hartree; a molecular mixed-response column agrees with the original implementation within 4.2×10^{-15} Hartree/bohr^2. The extended public surface and response are committed as `81897cf`; the exact contraction and metric acceleration is `e572d50`. Complete analytic Hessians are qualified for Criegee (two states, 15 Cartesian coordinates), furan (27 coordinates), and uracil (36 coordinates). Independently reconverged energies at 10^{-3} and 5×10^{-4} bohr validate both an out-of-plane direction and a mixed internal displacement. Richardson-extrapolated maximum errors are 1.42×10^{-10} Hartree/bohr for first derivatives and 3.71×10^{-7} Hartree/bohr^2 for curvatures. The largest raw Hessian asymmetry is 4.64×10^{-14} Hartree/bohr^2, and the maximum translational identity residual is 1.43×10^{-13} Hartree/bohr^2. These are implementation checks of the final anchored surface, not a validation of its equilibrium geometries or physical vibrational frequencies. Fresh-process runtime, spin and restart checks pass for all twenty states in eighteen molecules. Prepared energy-only evaluations take 1.7–9.0 s with two BLAS threads on MACSTUDIO, excluding model loading and reference preparation. They do not demonstrate a vertical speed advantage over TDA. A direct cold public-API Criegee Hessian reproduces the independently assembled matrices within 2.2×10^{-9} Hartree/bohr^2. Its complete two-state Hessian evaluation takes 418 s with two BLAS threads and about 1.0 GiB peak RSS; this is the current dense reference implementation, not an end-to-end timing advantage over TDA. The separate column-checkpoint route permits bounded remote parallel validation and reuses completed columns. The final LINK $\pi\pi^*$ test has restart error 3.6×10^{-15} Hartree and displaced trial overlap 0.99999996; trial evaluation leaves the accepted checkpoint unchanged.

5 Position, cost, and remaining validation

The rationale for the semiempirical representation is repeated molecular work with an explicit differentiable energy. Ground-engine primitives support orbital relaxation, coherent state tracking, gradients and nuclear curvature in the same prepared model. This complements the efficient multi-root response and optical-property goals of Grimme's simplified methods.[1, 3, 4] It does not establish an advantage for transition intensities or high-throughput multi-root spectroscopy.

TD-DFTB, TDDFT+TB and TDDFT-aas already offer important low-cost excited-state and gradient capabilities.[5, 7, 8] ΔDFTB supplies an especially close state-specific precedent, including molecular structures and vibrational properties.[9] OMx/ODMx–MRCI provides an established multireference semiempirical alternative.[10, 11] The contribution here is the explicit GFN2/DFT-response-informed energy, spin-pure orbital-optimized configurations, coherent state interaction, and full analytic response through the nuclear Hessian. It is not priority for semiempirical excited-state gradients. A documented TD-DFTB vibronic implementation obtains the excited-state Hessian by differences of analytic gradients;[23] this specific comparison is not a claim that analytic second derivatives are absent from every competing implementation.

The cost accounting is

$$T_{\text{workflow}} = T_{\text{reference}} + T_{\text{projection/screening}} + \sum_{k=1}^{N_{\text{geom}}} T_{\text{OO/CI/derivatives}}(R_k). \tag{11}$$

Reference preparation can be amortized, but its cost does not disappear. The dense complete AO-pair correction and frozen interaction can require fourth-power storage in the retained AO

dimension. The current single-root TDA vertical controls are faster than the prepared OO-CID evaluations; the measured advantage on this development set is accuracy and an explicit state-specific differentiated energy, not a demonstrated vertical speedup. An all-inclusive final-model size-scaling benchmark remains separate from the derivative-correctness checks. Earlier model-generation comparisons with published sTDA-xTB, molecular minima, frequencies and cost data are retained in the Supporting Information and are not attributed to this final model.

5.1 Prospective extension and anchor sensitivity

A final small extension fixed formaldehyde ($n\pi^*$), ethylene ($\pi\pi^*$) and bright butadiene (B_u, $\pi\pi^*$) before their new predictions. The 50/50 weights, exchange scale 0.9, screened interaction construction and energy origin were unchanged. All three predictions were saved before opening their QUEST numerical labels. A bibliographic search for structural references exposed an approximate formaldehyde vertical value after its prediction; the extension is therefore described as a prospective fixed-model test, rather than a strictly blinded experiment. A preparation amendment replaces a diffuse DFT-TDA seed projection by an explicit native valence promotion for ethylene and butadiene, followed by the same stationary KKT and state-fidelity checks. This removes a TDA requirement for these new state seeds; it does not change the energy model or fit their excitation energies. The completed formaldehyde prediction retains its original qualified reference transition.

Molecule	Reference	OO-CID	TDA	OO error
Formaldehyde	3.966	4.285	4.650	+0.319
Ethylene	7.897	9.079	11.430	+1.182
Butadiene	6.222	6.632	8.567	+0.410

The three-state OO-CID MAE is 0.637 eV, compared with 2.187 eV for the same-native-orbital TDA response control. OO-CID improves each of these controls, but ethylene remains 1.182 eV above its reference. The new test therefore does not sustain a universal 0.5 eV accuracy claim. In particular, ethylene's minimal valence chart and screened correlation treatment require further development; the present results do not isolate a single cause. The two-heavy-atom formaldehyde and ethylene anchors also activate the specified small frontier fallback because the current heavy-atom-plane certificate is unavailable. No case is removed or retrospectively refitted. Fresh public evaluations retain the assigned spin-pure states and pass restart and no-runtime-DFT/TDA/SCC checks.

Anchor tests displace one carbon by 0.01 and 0.03 bohr in an in-plane stretch or an out-of-plane direction, for the Criegee intermediate and formaldehyde. Each comparison uses the same physical final geometry, retains all originally prepared target states and reuses the completed reference checkpoints. With the original CSF membership transported by orbital overlap, the largest gap change is 0.047 eV for the Criegee $\pi\pi^*$ state and 0.030 eV for formaldehyde. These are local controls, not a bound on arbitrary anchor displacement. Re-selecting the configuration space at each new anchor is a different operation: loss of the Criegee reflection certificate reduces its space from 178 to 13 CSFs and changes the $\pi\pi^*$ gap by 0.263–0.273 eV. A trajectory must retain its frozen chart; symmetry-dependent space selection must not be silently reapplied at every displaced geometry. The implemented analytic surface already retains this membership during nuclear motion.

5.2 Predecessor structures, frequencies and exact acceleration

The frozen screened-even predecessor was optimized on the lowest singlet surfaces of formaldehyde and acetaldehyde using analytic gradients. Full analytic Hessians then assess the vibrational curvatures. Coupled-cluster internal coordinates provide a physical structural comparison,[24]

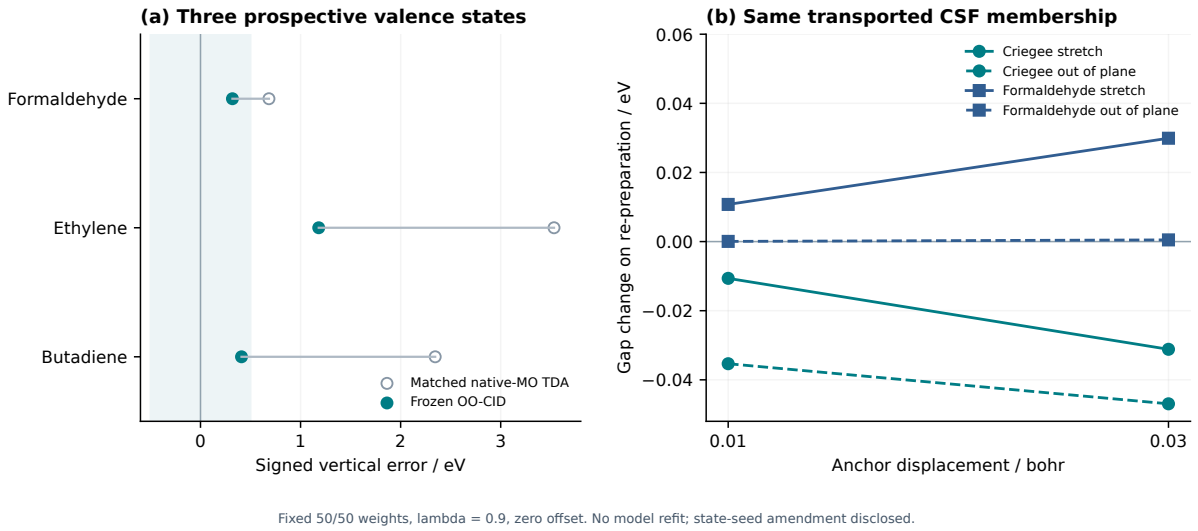


Figure 5: Final prospective extension and reference-geometry sensitivity. (a) OO-CID and matched native-orbital TDA signed errors for every new state; the shaded region is ±0.5 eV. (b) Re-prepared versus original-anchor excitation gaps evaluated at identical displaced geometries, with the original CSF membership transported by occupied/virtual orbital overlap. The model parameters are unchanged. The small extension is separate from the exposed twenty-state development set.

and selected experimental vibrational fundamentals provide a separately identified spectroscopic comparison.[25, 26] The model frequencies are unscaled harmonic curvatures, whereas the experimental numbers are anharmonic level spacings. Modes are assigned through projections on mass-weighted chemical-coordinate displacement patterns, rather than matched by frequency order. Inversion splittings and methyl-torsion level spacings are not used as harmonic curvature references. Complete coordinates, modes and individual differences are given in the Supporting Information. These comparisons assess that predecessor energy; they are not substituted for equilibrium data of the subsequently corrected reference.

A further exact response optimization vectorizes the overlap-residual completion and avoids scalar derivative-jet allocation for frozen auxiliary matrix values. A cold two-thread Criegee public-API Hessian takes 397 s, compared with the archived 418 s; energies, gradients and both full Hessians are bitwise identical between versions. Peak resident memory is 1.19 GB, compared with 1.06 GB. This single-run 4.9% time reduction is modest, with other work active on the host, and is not evidence of a size-scaling improvement or a speed advantage over TDA. The ten algebra, spin and legacy-contract regressions also pass. No numerical derivative, Hamiltonian change or fitted correction enters the acceleration.

5.3 Resolving the ethylene extension with a common screened reference

The prospective ethylene discrepancy motivates a second, separately identified development candidate. The F reference represents the full DFT Fock operator in a confined valence image, whereas its qualified Q levels come from the larger DFT calculation. To assess this imbalance, let P be the isometric native-valence image and Q its orthogonal complement in the complete DFT orbital space. For an already qualified orbital match, we evaluate the diagonal complement self-energy

$$\Sigma_i(E_i) = F_{iQ}(E_i - F_{QQ})^{-1}F_{Qi} \tag{12}$$

at its matched DFT orbital energy and add it to the F level before the unchanged 50/50 average. Unqualified matches retain their original reference. This is a diagonal downfolding approximation,

not an exact many-electron elimination of the DFT environment. No excitation reference enters this one-body preparation. It lowers the ethylene result from 9.079 to 8.206 eV, consistent with a substantial confinement contribution to the initial error.

The environment screening is rebuilt using Coulomb vertices between native occupied orbitals and an explicit orthogonal DFT bath, avoiding the use of missing active-orbital spectral weight as an independent screening channel. Subspace disentanglement is a recognized issue in constrained screening.[27] We allow one global semiempirical efficiency η for this additional bath polarization:

$$W = L_P^{\mathrm{T}} \left[I + B_N B_N^{\mathrm{T}} + \eta B_B B_B^{\mathrm{T}} \right]^{-1} L_P. \tag{13}$$

Here B_N contains the retained native environment transitions, B_B the occupied–bath transitions, and L_P the native Coulomb vertices. The one-body interaction is normal ordered to the corrected native ground Fock. The retained configuration space is unchanged. Static screening is an approximation whose quantitative limitations also arise in molecular RPA downfolding.[28]

We calibrate $\eta = 0.60$ by balancing the opposite errors of furan V010 and furofuran C002, and freeze it before evaluating the complete ledger. This is *one fitted interaction parameter*; it is neither an energy offset nor a parameter-free correction. It applies to every molecule and state. The F/Q weights stay 50/50, exchange-response λ stays 0.9, and orbital gaps are not multiplied by either parameter. All 23 numerical labels have been exposed during development, including the three-state extension, so this assessment is not a blind validation.

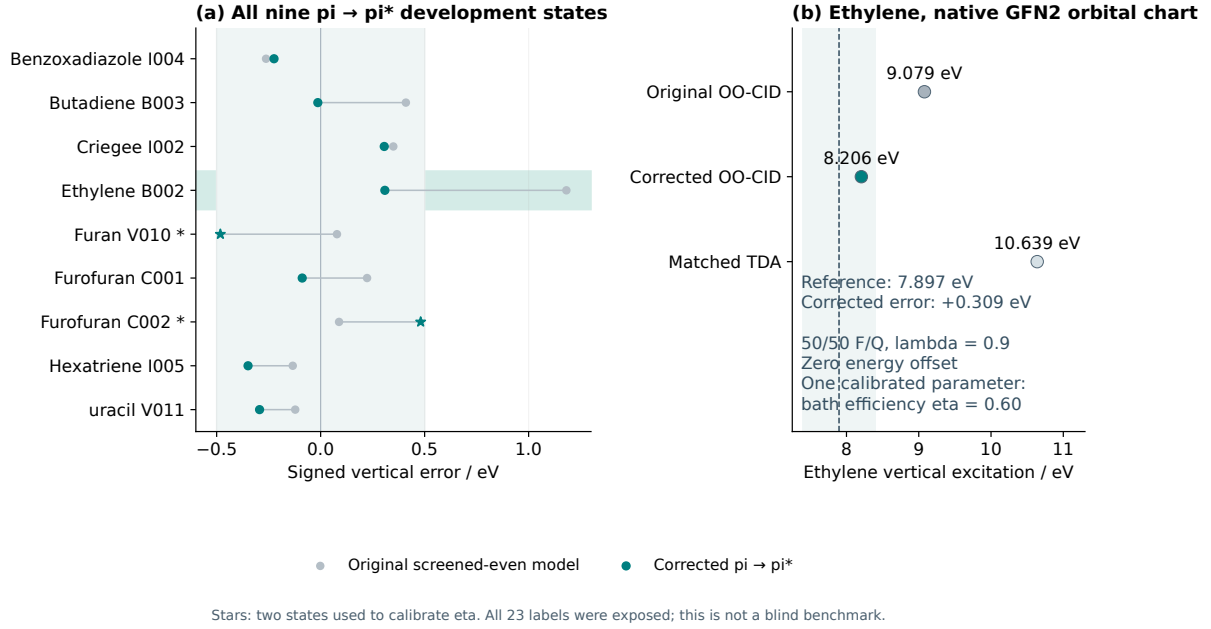


Figure 6: Common-reference repair on the complete exposed development set. (a) Original and corrected signed errors for all nine $\pi\pi^*$ states; stars identify the two states used to calibrate bath efficiency η. (b) Ethylene's initial and corrected OO-CID energies, with a freshly evaluated TDA control using the corrected native GFN2 ground orbitals and reference response. The reference is 7.897 eV. The shaded intervals are ±0.5 eV.

All nine $\pi\pi^*$ states now lie within 0.5 eV, with MAE 0.283 eV and maximum error 0.482 eV. The fourteen $n\pi^*$ states give 0.175 eV MAE and 0.381 eV maximum error. All 23 states give 0.217 eV MAE. The 21 states not used to choose η give 0.192 eV MAE; their earlier exposure still precludes a blind-validation claim. The matched corrected-reference TDA MAEs are 1.726 eV for $\pi\pi^*$, 0.174 eV for $n\pi^*$ and 0.781 eV overall. Thus the measured accuracy benefit is concentrated in $\pi\pi^*$; the two $n\pi^*$ mean errors are essentially equal. This response control is not a separately optimized published TDA parameterization.

Ethylene has error +0.309 eV. Its full analytic Hessian takes 10.15 s on the verified two-thread Mac Studio run; two independent scalar-energy Richardson checks give maximum errors 1.73×10^{-11} Ha/bohr for the gradient and 6.56×10^{-8} Ha/bohr^2 for directional curvature. The coherent singlet has $\langle S^2 \rangle = 0$, and its accepted restart agrees within 8.9×10^{-16} Ha. No DFT import, TDA solve or new SCC driver is used during prepared evaluation. These derivatives belong to the corrected frozen anchored surface. The earlier minimum and frequency comparisons in the preceding subsection belong to its screened-even predecessor and are not reassigned to this calibrated extension. Independent prospective molecules and corrected-model equilibrium structures remain necessary before a general transferability claim.

5.4 Fragmented orbital matches and three further valence tests

The fixed $\eta = 0.60$ candidate is next applied to cyclopropene, cyclopentadiene, and pyrrole. All three predictions are archived before extracting their numerical QUEST references. Their errors are initially +0.054, −0.013, and +2.369 eV. Historical exposure of some of these molecules prevents a strictly blind claim. Pyrrole's discrepancy exposes a reference-construction limitation: its native virtual orbitals have substantial weight distributed over several full-DFT virtual orbitals, so no individual match meets the existing 0.7 amplitude criterion. The original fallback therefore omits both the qualified Q level and the complement relaxation for these orbitals.

We amend this rule for *every* unresolved occupied or virtual orbital, without a molecule or state-family branch. Within the same occupation sector, full-DFT poles are ordered by native spectral weight w_{ki}. The smallest block $\mathcal{B}_i$ satisfying $\sum_{k\in\mathcal{B}_i} w_{ki} \geq 0.7^2$ supplies the normalized centroid

$$\mu_i = \frac{\sum_{k\in\mathcal{B}_i} w_{ki}\epsilon_k}{\sum_{k\in\mathcal{B}_i} w_{ki}}. \tag{14}$$

This centroid supplies the Q level with the original occupied-centroid alignment. Its difference from the Rayleigh energy of the original native image supplies the unresolved F relaxation. Individually qualified poles retain their preceding treatment. We then rebuild the explicit bath screening on the corrected native orbitals. The 50/50 average, $\lambda = 0.9$, $\eta = 0.60$, configuration rule and zero energy offset remain fixed. This is a diagonal spectral-centroid approximation, not an exact elimination of a broad or frequency-dependent environment. The reference preparer reads no excitation labels, but the amendment was developed after pyrrole scoring: the resulting ledger is an exposed development assessment.

Pyrrole then gives 6.302 eV against 6.247 eV. Applying the identical rule to all 26 states in 24 molecules gives MAE 0.198 eV and maximum error 0.446 eV. The twelve $\pi\pi^*$ states give 0.216 eV MAE; the fourteen $n\pi^*$ states give 0.183 eV. All individual errors are below 0.5 eV. Matched TDA on the corrected native GFN2 orbitals gives 0.789 eV MAE overall, 1.514 eV for $\pi\pi^*$ and 0.168 eV for $n\pi^*$: the OO gain is concentrated in the $\pi\pi^*$ family, while TDA has the smaller mean $n\pi^*$ error. Neither comparison is a calibrated sTDA-xTB benchmark. Every prepared state passes fresh-process spin and restart checks with no DFT import, new SCC call or TDA solve. Ethylene's updated full analytic Hessian takes 10.68 s; scalar-energy Richardson checks give errors below 9.92×10^{-12} Ha/bohr and 2.46×10^{-8} Ha/bohr^2. These remain derivatives of a frozen anchored model.

5.5 Matrix-free orbital response for larger valence spaces

For the pigment examples, the constrained ROKS orbital residual is solved by Newton–Krylov iterations. An analytic Jacobian action differentiates the native Fock and spin response, the orbital rotations, and the determinant ground-overlap constraint. The covariant orbital action is symmetric; the ground constraint is differentiated with singular-safe determinant cofactors. No dense orbital Jacobian is allocated and no finite difference enters the production orbital solver. All nonredundant orbital rotations are retained. A residual-decreasing line search controls each

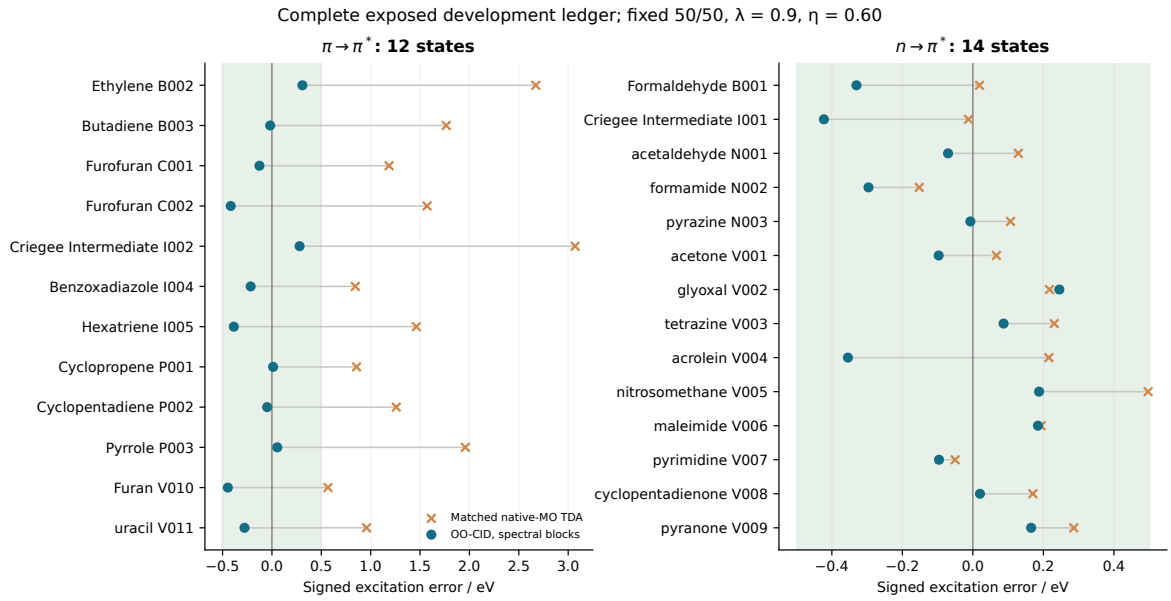


Figure 7: All 26 exposed development states after the generic spectral-block reference amendment. Points compare OO-CID with freshly calculated TDA on the same corrected native orbitals. The shaded interval is ±0.5 eV. The two panels use different horizontal scales. Neither the bath efficiency nor the F/Q weights are retuned for the three added molecules.

accepted step, while checkpointed overlap tracking follows the coherent singlet between accepted steps. Its character relative to the initial transition is reported separately. Final CI targets are assigned by collective overlap with the frozen native-frontier singlet subspace; this allows mixed frontier excitations without identifying a state by its energy. When a target fragments over several CI eigenstates, roots are retained until the smallest eigenvalue of their projector in the fixed target space reaches 0.49. This keeps the original coverage threshold, protects each target direction and is invariant to unitary rotations of the target basis.

The large-system XC response is evaluated on its full frozen quadrature. A density-fitted approximation preconditions the Newton linear system. When its step fails the full-response inexact-Newton tolerance, flexible GMRES refines it with analytic full-quadrature Hessian actions. The ground-constraint factorization is cached within each linear solve. For large Newton steps, a two-dimensional trust subspace spans the Newton and squared-residual descent directions. Minimizing the predicted residual in this subspace limits the orbital step without changing the stationarity equations; the scalar ground-constraint multiplier is optimized jointly. Writing $a = c, s$ for charge and spin channels, the actual energy is

$$E = E_{\mathrm{RI}} + \frac{1}{2}\sum_{a}\left[\int \delta\boldsymbol{\rho}_a(\mathbf{r})^{\mathsf{T}} f_{\mathrm{xc},a}(\mathbf{r})\delta\boldsymbol{\rho}_a(\mathbf{r})\,d\mathbf{r} - \mathbf{q}_a^{\mathsf{T}} K_{\mathrm{xc},a}^{\mathrm{RI}}\mathbf{q}_a\right], \tag{15}$$

where $\boldsymbol{\rho}$ contains the density and its three Cartesian derivatives. The charge deviation is measured from the anchored native ground density; the spin deviation is the spin density itself. Thus the approximate XC quadratic form is subtracted and the full quadrature form restored in both the energy and the convergence residual. Stored basis values, quadrature weights and XC kernels suffice for evaluation without a DFT backend. On ethylene, energy, Fock and response equivalence errors are below 2.5×10^{-15} in the corresponding atomic units; the spin-pure CI matrix agrees within 3.6×10^{-15} Ha. These controls distinguish an acceleration of the stationarity search from a change of energy model.

CI assembly likewise preserves the Hamiltonian while avoiding a full four-index molecular-orbital transformation: the common closed-shell core is contracted first and only the active

integral block is transformed. Fixed-bra Schur contractions reuse the common core in nonorthogonal couplings; repeated identical canonical occupations reuse their exact cofactors and pair contractions through a bounded cache. The canonical determinant-to-CSF transformation uses its exact sparse coefficients instead of dense multiplication; no determinant or spin coupling is dropped. Singular cofactors are evaluated as products of the remaining singular values without division. Registered density spans accelerate the quadratic native energy only when the requested density lies in that span; otherwise evaluation falls back to the full expression.

Native real d functions are included in the molecular-frame transport with analytic first and second rotation derivatives. Large-system native response also uses the same D4 and electrostatic cutoffs as its ground functional. Dense pair tensors and screened interactions are still stored: matrix-free orbital stationarity does not make this implementation linear-scaling or inexpensive in memory. The present quadrature adapter supports vertical energies and orbital response only; it explicitly rejects nuclear-derivative requests. The analytic nuclear Hessian qualifications reported for the small-molecule anchored surface therefore do not extend to these pigment calculations.

5.6 Larger pigment structures and a carotene vertical example

We select all-trans β-carotene and chlorophyllide a as larger chromophore tests. Chlorophyllide a retains Mg and the conjugated substituents while removing the phytyl chain. Figure 9 shows the actual source structures, rather than schematic molecular art. Native ground checkpoints contain 216 and 219 orbitals, respectively. The chlorophyllide reconstruction explicitly includes Mg d functions. Raw Gaussian-overlap discrepancies and subsequent numerical metric reconciliation are archived separately; agreement after reconciliation is 2.5×10^{-13} or better. The one-time PBE0/CAM-B3LYP reference preparation uses checkpointed RI-JK calculations on the verified Mac Studio. The carotene ROKS target converges at 3.518 eV, with orbital KKT residual 4.73×10^{-12} and ground-CSF overlap 1.32×10^{-13}. The initial 15-CSF screened-CID space incorrectly places an intense state at 3.269 eV below a weak state at 3.680 eV. We therefore expand the native frontier window with all spin-singlet configurations of excitation rank at most two, retaining the optimized ROKS configurations. The F/Q weights, λ, η, geometry and zero offset remain fixed. The resulting 15, 153, 351 and 703-CSF calculations are retained in the Supporting Information, including the initial ordering.

With five occupied and five virtual frontier orbitals, the first state becomes weak (2.561 eV, $f = 0.0238$), followed by the intense state (2.728 eV, $f = 3.127$). This initial ordering repair motivates further configuration-space checks rather than an absolute-energy convergence claim.

Larger fixed-screening calculations with $W = 10$ use 2,145, 3,403 and 5,151 CSFs for $N = 8, 9, 10$ (Figure 8). Their lowest weak state lies at 2.189, 2.145 and 2.084 eV, respectively. The frontier bright character fragments between roots 2 and 3: at $N = 9$ their individual native-target weights are 0.420 and 0.385, while their combined projector coverage is 0.805. Requiring a single root to carry 0.49 would therefore reject a resolved mixed target. The subspace criterion retains both roots with the same coverage threshold, without changing any matrix element or energy. Across $N = 9$ to 10, the squared principal cosines of the roots 2–3 subspace are 0.998 and 0.993, supporting its continuity through the intensity redistribution.

At $N = 10$, the lowest root has $f = 0.00883$; roots 2 and 3 lie at 2.467 and 2.532 eV with $f = 0.529$ and 2.310. Thus weak S_1 remains below the absorbing pair, but the dominant frontier intensity is now on the third energy-ordered root. Energy labels and the bright target must be distinguished through this mixing. The last increments of the three ordered gaps are -0.061, -0.058 and -0.041 eV. These do not establish absolute energy convergence. Screening-window effects are smaller: at fixed $N = 8$, increasing $W = 8$ to 10 changes the first two gaps by -0.009 and $+0.004$ eV. The complete sequence, transition strengths and overlap controls are given in the Supporting Information.

The first state is weak rather than strictly forbidden on the unoptimized source geometry. Room-temperature fluorescence-derived S_1 energies and solvent absorption maxima are different

observables and cannot validate these vertical gaps directly. The known low-lying dark state motivates the expanded configuration space [29]; pigment absorption data provide experimental context [30]. The expansion follows exposure of the spectra and is a development test, not blind validation. The chlorophyllide frontier calculations are still being qualified with serial checkpointed workers. Both pigments remain excluded from the 26-state accuracy pool; neither excited minima nor pigment nuclear Hessians are reported.

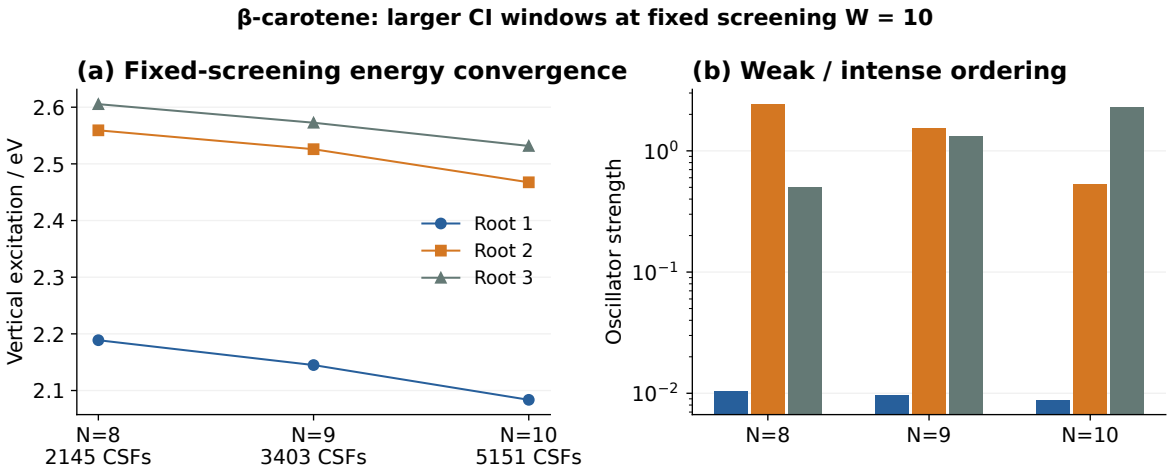


Figure 8: Fixed-screening configuration-window sensitivity of β-carotene at $W = 10$. The weak first singlet remains below an absorbing pair, whose intensity redistributes between roots 2 and 3. Lines connect energy-ordered roots and do not assert continuous diabatic identity. The Hamiltonian parameters and unoptimized source geometry remain unchanged. The initial 15–703 CSF ordering repair and the full window sequence are retained in the Supporting Information.

The next scientific assessment should test the calibrated correction on independent molecules, additional CT and Rydberg states, corrected-model equilibrium structures, and larger nuclear displacements. The prospective and structural controls reported here limit the scope of the finite development result; they do not establish universal transferability.

6 Conclusions

MOUSE provides a semiempirical orbital-optimized route with explicit spin polarization, spin-pure singlet CSFs, and analytic nuclear gradients and Hessians. Ground-program Fock, energy and derivative primitives are reused through constrained electronic saddle optimization and implicit response. The current candidate combines a spectral-block orbital reference and selected screened double excitations with fixed F/Q weights of 50/50, exchange-response scale 0.9 and one calibrated bath efficiency, $\eta = 0.60$. No excitation-energy offset is fitted. Across 26 exposed development states in 24 molecules it gives 0.198 eV MAE and 0.446 eV maximum error. The twelve $\pi\pi^*$ states give 0.216 eV MAE, compared with 1.514 eV for TDA on the same corrected native GFN2 orbitals. For the fourteen $n\pi^*$ states, TDA retains the smaller mean error, 0.168 versus 0.183 eV. These results establish an improvement on this finite development ledger, rather than a universal correction of either family.

The prepared surface retains coherent APOC state following and accepted-only LINK restarts without a runtime TDA solve, new SCC call or DFT-backend import. Complete implicit response differentiates the anchored energy, the moving ground projector and the state-interaction eigensystem. Independent energy checks support the analytic derivatives; the current ethylene full Hessian takes 10.68 s in one two-thread run. Molecular minima and physical frequency

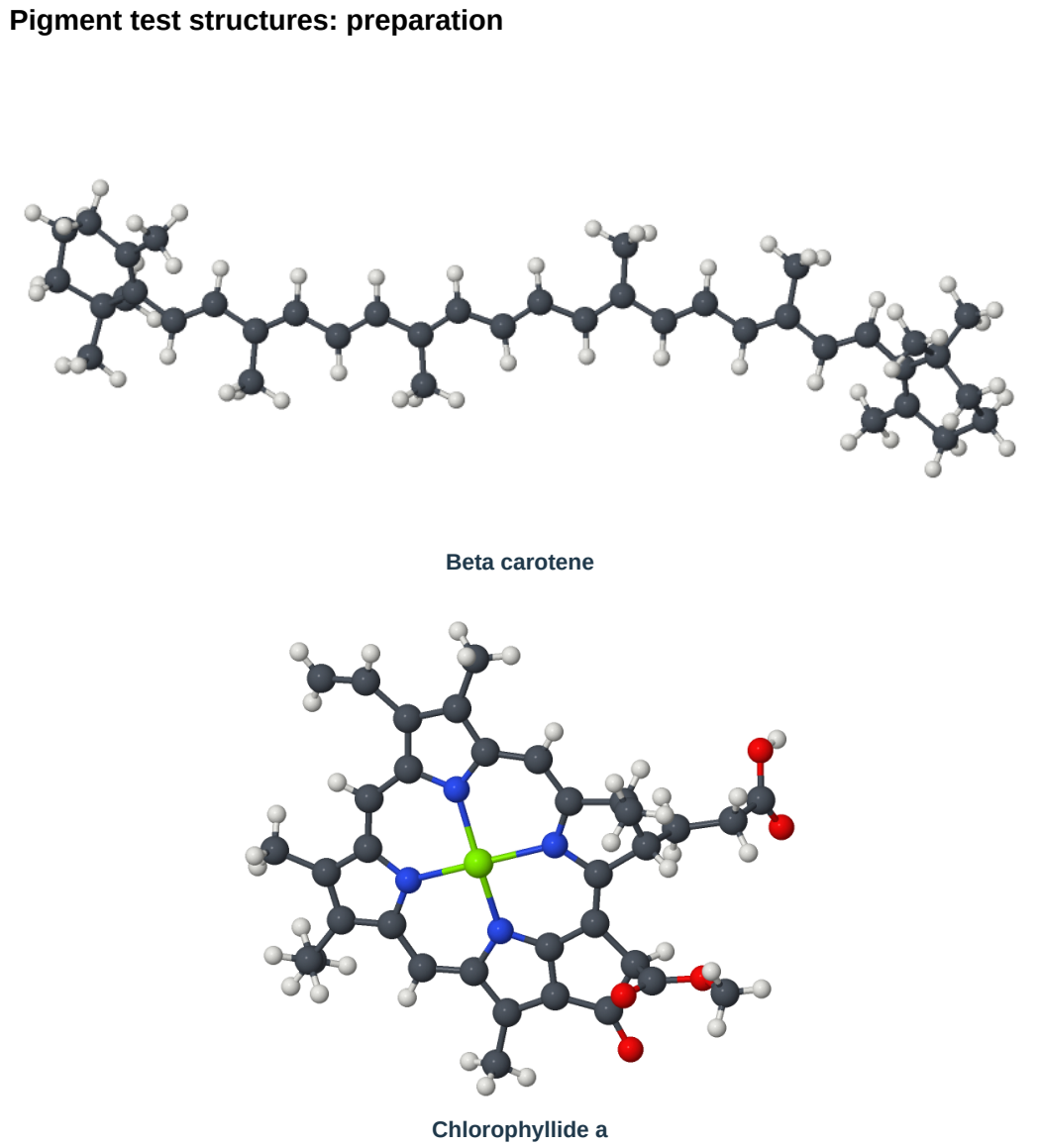


Figure 9: Jmol renderings of the larger starting structures: all-trans β-carotene (96 atoms; PubChem CID 5280489) and chlorophyllide a (79 atoms; RCSB ligand A1JWG). These primary-source geometries are used for preparation and are not optimized excited-state structures or validated additions to the accuracy ledger.

comparisons from earlier model generations remain separate from this corrected candidate's derivative qualifications.

Matrix-free orbital stationarity and exact frozen-quadrature XC correction extend the numerical implementation to larger native valence spaces. The carotene configuration expansion keeps a weak first singlet below the absorbing manifold without parameter adjustment. At 5,151 CSFs the first gap is 2.084 eV, and the dominant intensity is on the third ordered root at 2.532 eV after mixing with the second root. Collective target-space tracking resolves this fragmentation; absolute gaps remain sensitive to the CI window. Pigment predictions are excluded from the accuracy ledger, and their nuclear derivatives are not yet qualified. Full pair storage and DFT-reference preparation remain significant costs. Independent frozen-label validation, corrected-model equilibrium structures, broader CT/Rydberg and nonplanar cases, and matched calibrated sTDA-xTB/DFTB comparisons are required before claiming general predictive accuracy or replacement of multi-root TDA spectroscopy.

Data and software availability

MOUSE source is hosted at https://github.com/yogibubu/MOUSE. The final symmetry-augmentation choice is frozen in `2d59ab8`; the portable surface and analytic response are implemented in `81897cf`, with exact contraction/metric acceleration in `e572d50`. The closing protocol is frozen in `94d2312`, with exact dense response acceleration in `67c835f`. Its new cases, reference-geometry controls and molecular minima are archived in `qualification/closing_validation_20261101`. The calibrated bath protocol is frozen in `8f95105`; its complete 23-state ledger, matched corrected-reference TDA and ethylene derivative certificate are in `qualification/closing_validation_20261101`. Original twenty-state qualification and its complete logs are committed in `358ae21` and `7c5c7f9`. The qualification directory `qualification/pi_pi_transfer_repair_20261101` contains the complete twenty-state predictions, model/exposure protocol, remote runtime and spin checks, analytic-column checkpoints, independent energy validations and executed-source hashes. The manuscript package contains the machine-readable results, scientific figure sources, archived Jmol renders and coordinate provenance. Earlier generations and their qualification archives are identified separately in the Supporting Information and preserved development manuscript. The large-system runtime and exact accelerations are archived in `d05afa6` and `e3c005a`, with fragmented-target tracking in `bb3435f`. Large pigment checkpoints remain on the verified compute host with checksummed receipts and compact low-state exports. A permanent public dataset identifier has not yet been assigned.

References

[1] Stefan Grimme. A simplified Tamm-Dancoff density functional approach for the electronic excitation spectra of very large molecules. *The Journal of Chemical Physics*, 138:244104, 2013. doi: 10.1063/1.4811331.

[2] Christoph Bannwarth and Stefan Grimme. A simplified time-dependent density functional theory approach for electronic ultraviolet and circular dichroism spectra of very large molecules. *Computational and Theoretical Chemistry*, 1040-1041:45–53, 2014. doi: 10.1016/j.comptc.2014.02.023.

[3] Stefan Grimme and Christoph Bannwarth. Ultra-fast computation of electronic spectra for large systems by tight-binding based simplified Tamm-Dancoff approximation (sTDA-xTB). *The Journal of Chemical Physics*, 145:054103, 2016. doi: 10.1063/1.4959605.

[4] Pierre Beaujean, Axel Liger, Arya Satisha, Marilù G. Maraldi, Christoph Bannwarth, and Marc de Wergifosse. The std2 package: A computationally efficient program to calculate

molecular response properties and excited states using simplified quantum chemistry methods. *The Journal of Chemical Physics*, 165:152501, 2026. doi: 10.1063/5.0349975.

[5] B. Hourahine, M. Berdakin, J. A. Bich, F. P. Bonafé, C. Camacho, Q. Cui, M. Y. Deshaye, G. Díaz Mirón, S. Ehlert, M. Elstner, T. Frauenheim, N. Goldman, R. A. González León, T. van der Heide, S. Irle, T. Kowalczyk, T. Kubař, I. S. Lee, C. R. Lien-Medrano, A. Maryewski, T. Melson, S. K. Min, T. Niehaus, A. M. N. Niklasson, A. Pecchia, K. Reuter, C. G. Sánchez, C. Scheurer, M. A. Sentef, P. V. Stishenko, V. Q. Vuong, and B. Aradi. Recent Developments in DFTB+, a Software Package for Efficient Atomistic Quantum Mechanical Simulations. *The Journal of Physical Chemistry A*, 129:5373–5390, 2025. doi: 10.1021/acs.jpca.5c01146.

[6] Wenbo Sun, Tammo van der Heide, Van-Quan Vuong, Thomas Frauenheim, Michael A. Sentef, Bálint Aradi, and Carlos R. Lien-Medrano. Hybrid Functional DFTB Parametrizations for Modeling Organic Photovoltaic Systems. *Journal of Chemical Theory and Computation*, 21:5103–5117, 2025. doi: 10.1021/acs.jctc.5c00232.

[7] Shana Havenridge, Robert Rüger, and Christine M. Aikens. Analytical excited state gradients for time-dependent density functional theory plus tight binding (TDDFT + TB). *The Journal of Chemical Physics*, 158:224103, 2023. doi: 10.1063/5.0142240.

[8] Yuchen Wang, Shana Havenridge, and Christine M. Aikens. Implementation of energy and gradient for the TDDFT-approximate auxiliary function (aas) method. *The Journal of Chemical Physics*, 161:024101, 2024. doi: 10.1063/5.0213587.

[9] Tim Kowalczyk, Khoa Le, and Stephan Irle. Self-Consistent Optimization of Excited States within Density-Functional Tight-Binding. *Journal of Chemical Theory and Computation*, 12:313–323, 2016. doi: 10.1021/acs.jctc.5b00734.

[10] Deniz Tuna, You Lu, Axel Koslowski, and Walter Thiel. Semiempirical Quantum-Chemical Orthogonalization-Corrected Methods: Benchmarks of Electronically Excited States. *Journal of Chemical Theory and Computation*, 12:4400–4422, 2016. doi: 10.1021/acs.jctc.6b00403.

[11] Pavlo O. Dral, Xin Wu, and Walter Thiel. Semiempirical Quantum-Chemical Methods with Orthogonalization and Dispersion Corrections. *Journal of Chemical Theory and Computation*, 15:1743–1760, 2019. doi: 10.1021/acs.jctc.8b01265.

[12] Gianluca Levi, Aleksei V. Ivanov, and Hannes Jónsson. Variational calculations of excited states via direct optimization of the orbitals in DFT. *Faraday Discussions*, 224:448–466, 2020. doi: 10.1039/D0FD00064G.

[13] Gianluca Levi, Aleksei V. Ivanov, and Hannes Jónsson. Variational Density Functional Calculations of Excited States via Direct Optimization. *Journal of Chemical Theory and Computation*, 16:6968–6982, 2020. doi: 10.1021/acs.jctc.0c00597.

[14] Aleksei V. Ivanov, Gianluca Levi, Elvar Ö. Jónsson, and Hannes Jónsson. Method for Calculating Excited Electronic States Using Density Functionals and Direct Orbital Optimization with Real Space Grid or Plane-Wave Basis Set. *Journal of Chemical Theory and Computation*, 17:5034–5049, 2021. doi: 10.1021/acs.jctc.1c00157.

[15] Yorick L. A. Schmerwitz, Gianluca Levi, and Hannes Jónsson. Calculations of Excited Electronic States by Converging on Saddle Points Using Generalized Mode Following. *Journal of Chemical Theory and Computation*, 19:3634–3651, 2023. doi: 10.1021/acs.jctc.3c00178.

[16] Yorick L. A. Schmerwitz, Elli Selenius, and Gianluca Levi. Freeze-and-Release Direct Optimization Method for Variational Calculations of Excited Electronic States. *Journal of Chemical Theory and Computation*, 22:3571–3584, 2026. doi: 10.1021/acs.jctc.5c01974.

[17] Christoph Bannwarth, Sebastian Ehlert, and Stefan Grimme. GFN2-xTB: An Accurate and Broadly Parametrized Self-Consistent Tight-Binding Quantum Chemical Method with Multipole Electrostatics and Density-Dependent Dispersion Contributions. *Journal of Chemical Theory and Computation*, 15:1652–1671, 2019. doi: 10.1021/acs.jctc.8b01176.

[18] Irmgard Frank, Jürg Hutter, Dominik Marx, and Michele Parrinello. Molecular dynamics in low-spin excited states. *The Journal of Chemical Physics*, 108:4060–4069, 1998. doi: 10.1063/1.475804.

[19] Takeshi Yanai, David P. Tew, and Nicholas C. Handy. A new hybrid exchange–correlation functional using the coulomb-attenuating method (CAM-B3LYP). *Chemical Physics Letters*, 393:51–57, 2004. doi: 10.1016/j.cplett.2004.06.011.

[20] Mickaël Véril, Anthony Scemama, Michel Caffarel, Filippo Lipparini, Martial Boggio-Pasqua, Denis Jacquemin, and Pierre-François Loos. QUESTDB: A database of highly accurate excitation energies for the electronic structure community. *WIREs Computational Molecular Science*, 11:e1517, 2021. doi: 10.1002/wcms.1517.

[21] Pierre-François Loos, Martial Boggio-Pasqua, Aymeric Blondel, Filippo Lipparini, and Denis Jacquemin. QUEST Database of Highly-Accurate Excitation Energies. *Journal of Chemical Theory and Computation*, 21:8010–8033, 2025. doi: 10.1021/acs.jctc.5c00975.

[22] F. Aryasetiawan, M. Imada, A. Georges, G. Kotliar, S. Biermann, and A. I. Lichtenstein. Frequency-dependent local interactions and low-energy effective models from electronic structure calculations. *Physical Review B*, 70:195104, 2004. doi: 10.1103/PhysRevB.70.195104.

[23] Robert Rüger, Thomas Niehaus, Erik van Lenthe, Thomas Heine, and Lucas Visscher. Vibrationally resolved UV/Vis spectroscopy with time-dependent density functional based tight binding. *The Journal of Chemical Physics*, 145:184102, 2016. doi: 10.1063/1.4966918.

[24] Šimon Budzák, Giovanni Scalmani, and Denis Jacquemin. Accurate Excited-State Geometries: A CASPT2 and Coupled-Cluster Reference Database for Small Molecules. *Journal of Chemical Theory and Computation*, 13:6237–6252, 2017. doi: 10.1021/acs.jctc.7b00921.

[25] V. A. Job, V. Sethuraman, and K. K. Innes. The 3500 å transition of formaldehyde-h2, d2, and hd: Vibrational and rotational analyses. *Journal of Molecular Spectroscopy*, 30:365–426, 1969. doi: 10.1016/0022-2852(69)90274-4.

[26] Lynn M. Hubbard, David F. Bocian, and Robert R. Birge. The nature of the $^1n\pi^* \leftarrow S_0$ transition. 4. the first excited singlet state of acetaldehyde. *Journal of the American Chemical Society*, 103:3313–3320, 1981. doi: 10.1021/ja00402a016.

[27] Takashi Miyake, Ferdi Aryasetiawan, and Masatoshi Imada. Ab initio procedure for constructing effective models of correlated materials with entangled band structure. *Physical Review B*, 80:155134, 2009. doi: 10.1103/PhysRevB.80.155134.

[28] Erik Verzijl and Arno Förster. Static effective hamiltonians for molecular systems through RPA-based downfolding. *Journal of Chemical Theory and Computation*, 22(18):9468–9482, 2026. doi: 10.1021/acs.jctc.6c01128.

[29] S. L. Bondarev and V. N. Knyukshto. Fluorescence from the S1 state of all-trans-beta-carotene. *Chemical Physics Letters*, 225:346–350, 1994. doi: 10.1016/0009-2614(94)87092-6.

[30] Lesley A. Clementson and Bozena Wojtasiewicz. Dataset on the absorption characteristics of extracted phytoplankton pigments. *Data in Brief*, 24:103875, 2019. doi: 10.1016/j.dib.2019.103875.